Project Report: Customer Churn Classifier with MLflow - ML / AI Engineer

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Family:	ML / AI Engineer
Level:	Intermediate
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

Classification pipeline logged in MLflow, compared experiments, deployed as FastAPI endpoint.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Intermediate]

Chapter focus: Project Overview *(Level: Intermediate)**

Customer Churn Classifier with MLflow is a intermediate-level end-to-end project for ML / AI Engineer. Duration: 3-4 weeks. Classification pipeline logged in MLflow, compared experiments, deployed as FastAPI endpoint. Tools: scikit-learn, MLflow, FastAPI, SHAP.**

Code Reference:

```
Objective: Feature engineering for churn
Objective: Train 3 models and compare in MLflow
Objective: Select best by F1/recall
Objective: Deploy as REST API
```

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is
A project overview defines scope, tools, and success criteria.

When to use it
At project kickoff.

Where to use it
ML / AI Engineer portfolios and CodeQuest project submissions.

Real use example
A candidate lists this project on a ML / AI Engineer resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: Data [Intermediate]

Chapter focus: Phase: Data *(Level: Intermediate)*

Feature engineering. Complete these tasks: RFM features; Train/test split. Deliverable for this phase: Feature matrix.

Code Reference:

```
- [ ] RFM features
- [ ] Train/test split
```

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is
Phase Data is one step in the full project lifecycle.

When to use it
During week 1 of the project timeline.

Where to use it
ML / AI Engineer agile sprints and coursework milestones.

Real use example
Team demo shows the Feature matrix to the teacher.

Chapter 3: Phase: Experiments [Intermediate]

Chapter focus: Phase: Experiments *(Level: Intermediate)*

MLflow tracking. Complete these tasks: 3 algorithms; Parameter logging. Deliverable for this phase: MLflow UI screenshot.

Code Reference:

```
- [ ] 3 algorithms
- [ ] Parameter logging
```

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is
Phase Experiments is one step in the full project lifecycle.

When to use it
During week 2 of the project timeline.

Where to use it
ML / AI Engineer agile sprints and coursework milestones.

Real use example
Team demo shows the MLflow UI screenshot to the teacher.

Chapter 4: Phase: Select [Intermediate]

Chapter focus: Phase: Select *(Level: Intermediate)**

Best model by recall. Complete these tasks: Confusion matrix; SHAP summary. Deliverable for this phase: Model comparison report.

Code Reference:

```
- [ ] Confusion matrix
- [ ] SHAP summary
```

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is
Phase Select is one step in the full project lifecycle.

When to use it
During week 3 of the project timeline.

Where to use it
ML / AI Engineer agile sprints and coursework milestones.

Real use example
Team demo shows the Model comparison report to the teacher.

Chapter 5: Phase: Serve [Intermediate]

Chapter focus: Phase: Serve *(Level: Intermediate)**

FastAPI endpoint. Complete these tasks: /predict; Input schema. Deliverable for this phase: Live prediction API.

Code Reference:

```
- [ ] /predict
- [ ] Input schema
```

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is
Phase Serve is one step in the full project lifecycle.

When to use it
During week 4 of the project timeline.

Where to use it
ML / AI Engineer agile sprints and coursework milestones.

Real use example
Team demo shows the Live prediction API to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Intermediate]

Chapter focus: Final Deliverables and Review *(Level: Intermediate)**

Submit all final deliverables: MLflow experiments; Best model; API endpoint; Model card. Skills demonstrated: Classification, MLflow, Deployment. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

```
Deliverable: MLflow experiments
Deliverable: Best model
Deliverable: API endpoint
Deliverable: Model card
```

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide

What it actually is
Deliverable review confirms end-to-end project completion.

When to use it
Before deadline and teacher review.

Where to use it
CodeQuest project gallery and interviews.

Real use example
A student submits MLflow experiments and passes the rubric.